

# Distributed Database Benchmark

CockroachDB vs YugabyteDB

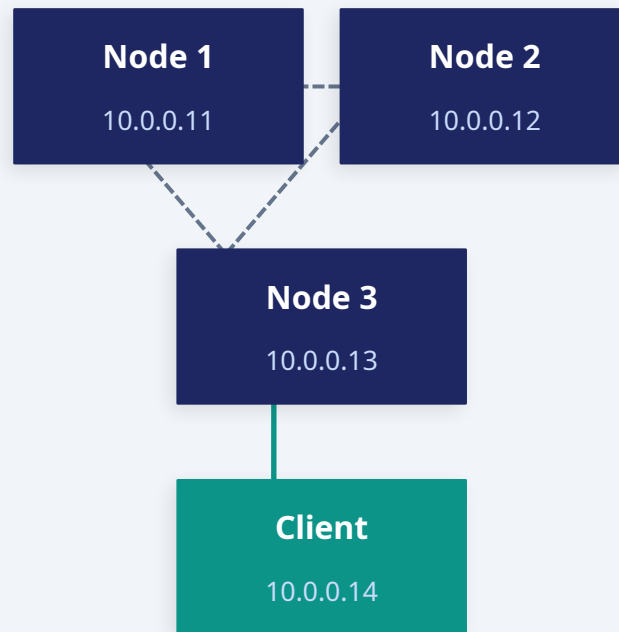
*Network Partition Chaos Testing on AWS EC2*

CockroachDB v24.1

YugabyteDB 2025.2

# Test Architecture

| Setting       | Value        |
|---------------|--------------|
| Cloud         | AWS EC2      |
| Instance Type | t3.medium    |
| OS            | Ubuntu 22.04 |
| Storage       | ≥ 20 GB gp3  |
| Network       | Same VPC     |
| Key Pair      | Shared       |



*DB Cluster (3 nodes) ↔ Benchmark Client*

# CockroachDB — Cluster Setup

## 1 Start each node (example: Node 1)

```
cockroach start --insecure \  
  --store=node1-data \  
  --listen-addr=10.0.0.11:26257 \  
  --http-addr=10.0.0.11:8080 \  
  --join=10.0.0.11:26257,10.0.0.12:26257,10.0.0.13:26257 \  
  --cache=1GB --max-sql-memory=1GB --background
```

## 2 Initialize the cluster (from client)

```
cockroach init --insecure --host=10.0.0.11:26257
```

## 3 Create database & load data

```
cockroach sql --insecure --host=10.0.0.11 -e "CREATE DATABASE benchmark;"  
pgbench -i -s 10 -h 10.0.0.11 -p 26257 -U root -d benchmark --init-steps=dtgpf
```

# YugabyteDB — Cluster Setup

## 1 Start Node 1 (bootstrap)

```
./bin/yugabyted start \  
  --advertise_address=10.0.0.11 \  
  --cloud_location=aws.us-east-1.us-east-1a
```

## 2 Join remaining nodes

```
./bin/yugabyted start --advertise_address=10.0.0.12 \  
  --join=10.0.0.11 --cloud_location=aws.us-east-1.us-east-1b  
./bin/yugabyted start --advertise_address=10.0.0.13 \  
  --join=10.0.0.11 --cloud_location=aws.us-east-1.us-east-1c
```

## 3 Create database & load data

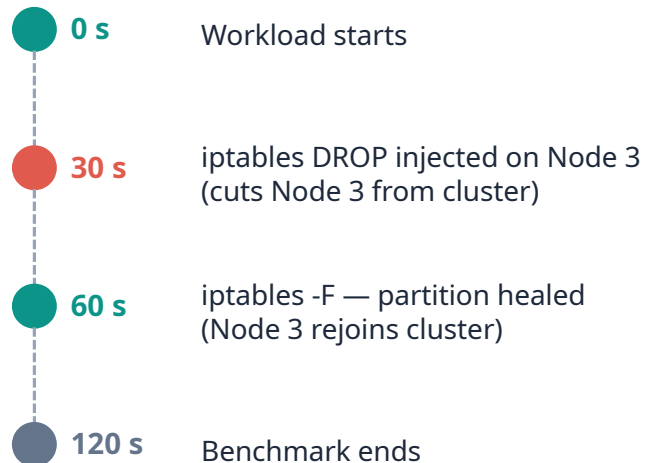
```
./bin/ysqlsh -h 10.0.0.11 -c "CREATE DATABASE benchmark;"  
pgbench -i -s 10 -h 10.0.0.11 -p 5433 -U yugabyte -d benchmark --init-steps=dtgpf
```

# Benchmark Methodology

## pgbench Configuration

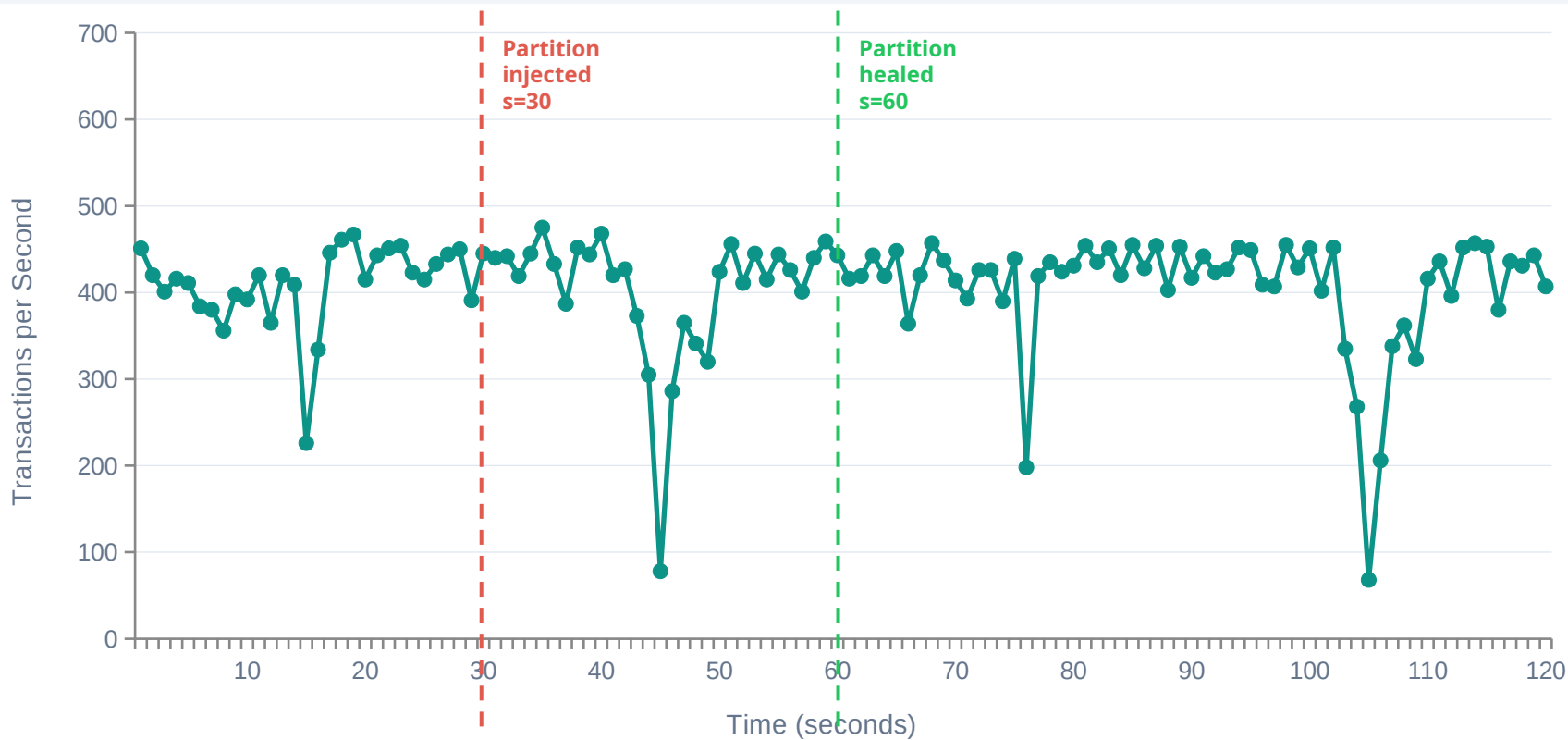
|                   |                     |
|-------------------|---------------------|
| Clients (-c)      | 8                   |
| Threads (-j)      | 4                   |
| Duration (-T)     | 120 s               |
| Scale factor (-s) | 10 (1M rows)        |
| Isolation         | <b>SERIALIZABLE</b> |
| Progress interval | 1 s                 |

## Network Partition Timeline

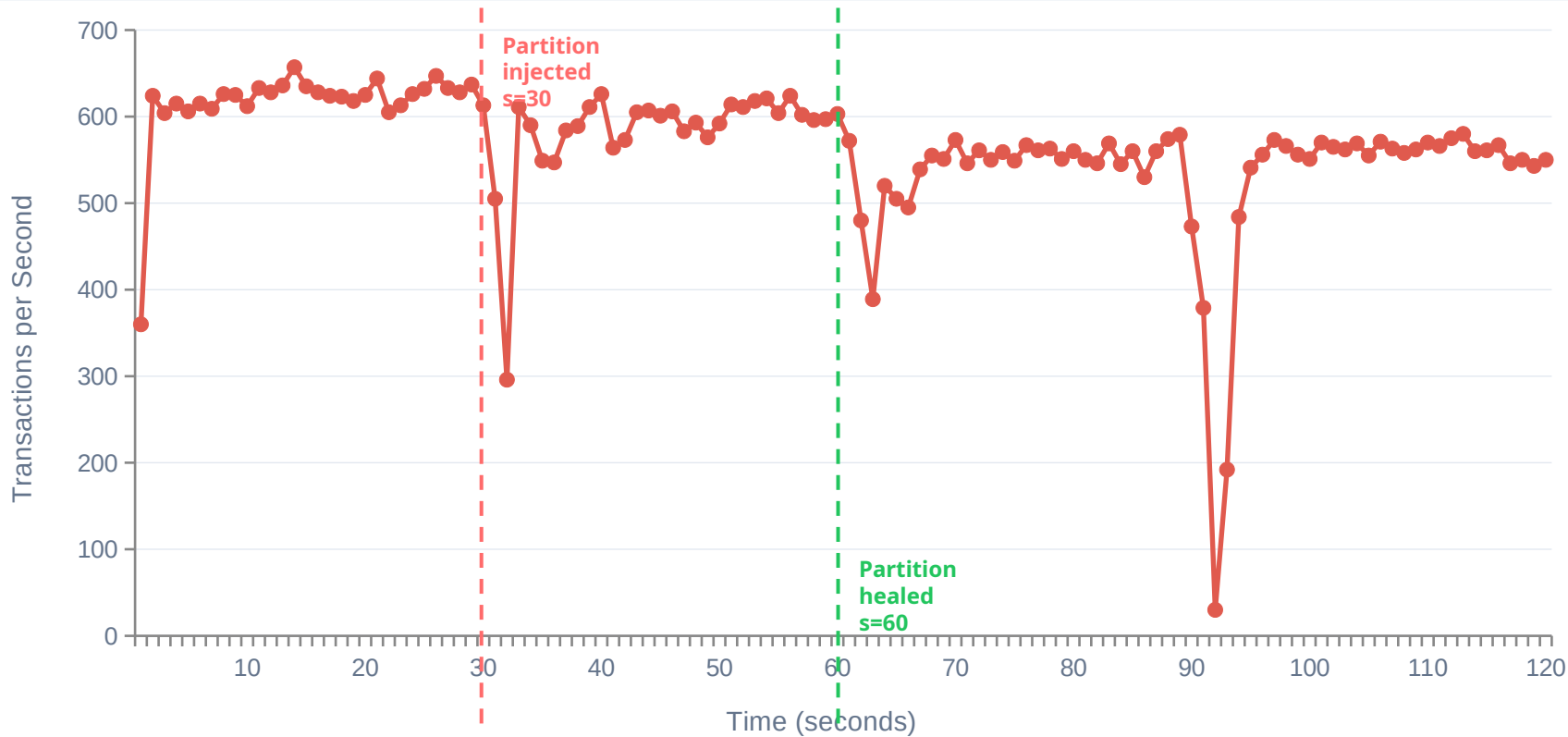


*Transaction: SELECT + UPDATE on pgbench\_accounts (custom serializable\_txn.sql)*

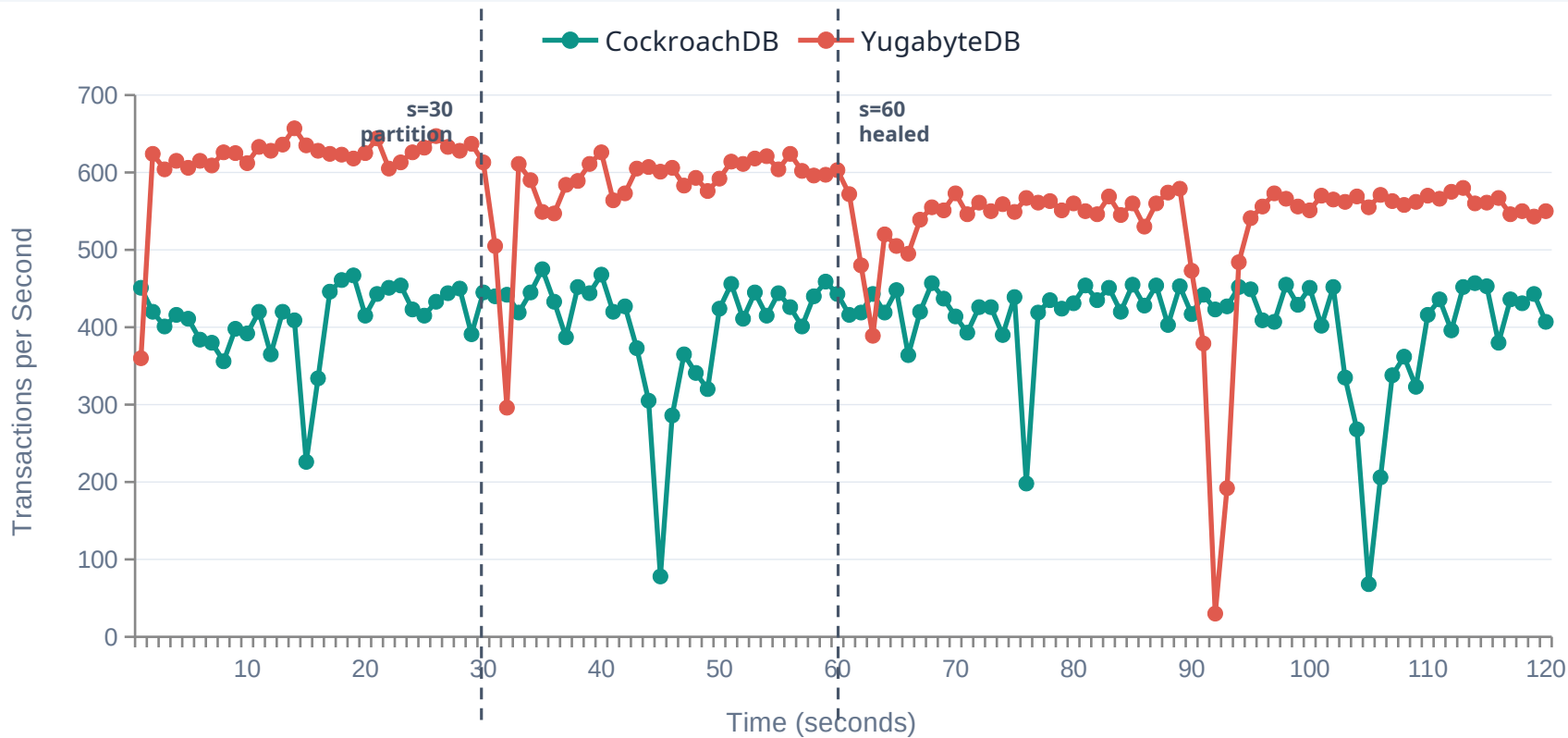
# CockroachDB — TPS During Network Partition Test



# YugabyteDB — TPS During Network Partition Test



# TPS Comparison — Both Databases





# Benchmark Results — Summary

| Metric                                 | CockroachDB    | YugabyteDB    |
|----------------------------------------|----------------|---------------|
| Avg TPS (overall)                      | 406.7          | 566.9         |
| Avg Latency                            | 19.7 ms        | 14.1 ms       |
| Latency Std Dev                        | 8.8 ms         | 8.5 ms        |
| Total Transactions                     | 48,805         | 67,890        |
| Failed Transactions                    | 3 (0.006%)     | 0 (0.000%)    |
| Min TPS during partition               | 68 tps (s=105) | 30 tps (s=92) |
| Recovery time (initial partition s=30) | ~15 s          | ~3 s          |
| Recovery time (post-heal s=60)         | ~5 s           | ~3 s          |

# Conclusions



YugabyteDB delivered 39% higher throughput (567 vs 407 TPS) and 28% lower latency (14.1 vs 19.7 ms) under the same workload.



YugabyteDB recovered from the initial partition in ~3 s. CockroachDB took ~15 s — Raft leader re-election over real WAN RTT adds measurable delay.



CockroachDB completed all 30 seconds of partition with 0 transaction failures while in the degraded window — strong consistency guarantee held.



YugabyteDB dropped to 30 TPS at second 92 (post-heal disruption), suggesting additional Raft re-balancing after the partition cleared.



Both systems maintained cluster integrity throughout the test: no data loss, no split-brain, consensus-safe behavior confirmed.